TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	56.8 ± 3.16
Correlation Stability	80.2 ± 1.40
Distribution Consistency	71.0 ± 8.64
Entropy Preservation	-71.4 ± 11.37
Mean Z Score Drift	0.6 ± 0.04
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	2.3 ± 1.55
Data Density	100.0 ± 0.00
Coverage Score	93.6 ± 3.23
Feature Accuracy	65.3 ± 1.49
Novelty Score	0.7 ± 0.04

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Dataset Audit: hepatitis.csv

1. Dataset Portrait

Attribute	Value
Rows	155
Columns	20
Numeric features	20
Categorical features	0
Missing cells (%)	5.4%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	41.200	45.743	+4.543	11.0%	Warn
Sex	1.103	1.000	-0.103	9.4%	OK
Steroid	1.506	1.279	-0.227	15.1%	Warn
Antivirals	1.845	1.919	+0.074	4.0%	OK
Fatigue	1.351	1.051	-0.300	22.2%	Warn
Malaise	1.604	1.264	-0.340	21.2%	Warn
Anorexia	1.792	1.615	-0.177	9.9%	OK
Liver Big	1.828	1.941	+0.114	6.2%	OK
Liver Firm	1.583	1.753	+0.170	10.7%	Warn
Spleen Palpable	1.800	1.530	-0.270	15.0%	Warn
Spiders	1.660	1.261	-0.399	24.0%	Warn
Ascites	1.867	1.561	-0.306	16.4%	Warn

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	56.8 ± 3.16
Correlation Stability (*2)	80.2 ± 1.40
Distribution Consistency (*3)	71.0 ± 8.64
Entropy Preservation (*4)	-71.4 ± 11.37
Mean Z Score Drift (*5)	0.6 ± 0.04
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	2.3 ± 1.55

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	93.6 ± 3.23
Feature Accuracy (*10)	65.3 ± 1.49
Novelty Score (*11)	0.7 ± 0.04

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8064516129032258	N/A	0.5290322580645161	-0.27741935483870966	*
F1	N/A	N/A	0.3431746593125903 4	0.34317465931259034	*

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8387096774193548	N/A	0.8258064516129032	-0.012903225806451623	
F1	0.5718392134181608	N/A	0.5870588235294119	0.01521961011125098	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8548387096774194	N/A	0.8451612903225806	-0.009677419354838712	
F1	0.5877844377844378	N/A	0.5938976572103508	0.006113219425912947	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7806451612903226	N/A	0.4999999999999999	-0.2806451612903226	*
F1	N/A	N/A	0.2601559138801702	0.2601559138801702	*

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8096774193548386	N/A	0.7193548387096774	-0.09032258064516126	
F1	0.4338278388278389	N/A	0.4334228942413086	-0.00040494458653024 81	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7806451612903226	N/A	0.8129032258064516	0.03225806451612904	
F1	0.4639843489843489	N/A	0.4810344041922990 4	0.01705005520794993	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8580645161290322	N/A	0.861290322580645	0.00322580645161287 8	
F1	0.5932576195734091	N/A	0.6201035239270533	0.02684590435364428 4	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7903225806451613	N/A	0.7774193548387097	-0.01290322580645162 3	
F1	0.561449871078354	N/A	0.5429623253983868	-0.01848754567996725	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8387096774193548	N/A	0.8032258064516128	-0.03548387096774193 6	
F1	0.5132478632478632	N/A	0.5409913615795968	0.02774349833173363	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8516129032258064	N/A	0.832258064516129	-0.01935483870967740 3	
F1	0.5385593354014405	N/A	0.5392486907657186	0.00068935536427796 6	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	30.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0
2	50.0	1.0	1.0	2.0	1.0	2.0	2.0	1.0	2.0	2.0
3	78.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
4	31.0	1.0	nan	1.0	2.0	2.0	2.0	2.0	2.0	2.0
5	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
6	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
7	51.0	1.0	1.0	2.0	1.0	2.0	1.0	2.0	2.0	1.0
8	23.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
9	39.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
10	30.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
11	39.0	1.0	1.0	1.0	2.0	2.0	2.0	1.0	1.0	2.0
12	32.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	1.0	2.0
13	41.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	1.0	2.0
14	30.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
15	47.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
16	38.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
17	66.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
18	40.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
19	38.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
20	38.0	1.0	1.0	1.0	2.0	2.0	2.0	1.0	1.0	2.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	57.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
2	52.0	1.0	1.0	2.0	1.0	1.819085404742099	1.180914595257901	2.0	2.0	1.180914595257901
3	35.0	1.0	1.0	2.0	1.0	1.9530402209867415	1.9530402209867415	2.0	1.9530402209867415	1.0469597790132585
4	43.0	1.0	1.0	2.0	1.0	1.0	1.5880470166136413	1.0	1.0	2.0
5	50.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	1.1349828793928183
6	31.0	1.0	1.6721772794761716	2.0	1.0	1.3278227205238324	1.3278227205238324	2.0	1.3278227205238324	1.6721772794761716
7	61.0	1.0	1.0	2.0	1.0	1.0	2.0	nan	nan	2.0
8	23.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
9	30.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	1.0	2.0
10	30.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	44.0	1.0	1.9411010786279226	2.0	1.0	1.0	1.0	2.0	2.0	1.9411010786279226
2	58.0	1.0	1.0	2.0	1.0	1.2889043137969713	1.7110956862030287	2.0	2.0	1.7110956862030287
3	54.0	1.0	1.0	2.0	1.0	1.0	1.0	1.0725078133976362	1.0725078133976362	1.9274921866023638
4	45.0	1.0	2.0	2.0	1.8826115819298797	2.0	2.0	2.0	2.0	2.0
5	48.0	1.0	1.7260578185665585	2.0	1.7260578185665585	1.7260578185665585	2.0	2.0	2.0	1.7260578185665585
6	47.0	1.0	2.0	2.0	1.0	1.1666337875582444	1.1666337875582444	1.8333662124417556	1.8333662124417556	1.8333662124417556
7	60.0	1.0	1.0	2.0	1.0	2.0	2.0	1.0	1.0	1.0
8	39.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
9	30.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	36.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	60.0	1.0	1.0	2.0	1.0	1.0777220225422344	1.9222779774577656	2.0	2.0	1.9222779774577656
2	33.0	1.0	1.3336675287765325	2.0	1.0	1.6663324712234675	1.6663324712234675	2.0	1.6663324712234675	1.3336675287765325
3	49.0	1.0	2.0	2.0	1.0	1.7449432446631967	2.0	2.0	1.0	1.2550567553368033
4	33.0	1.0	1.7486837471621484	2.0	1.0	1.0	1.2513162528378516	2.0	1.2513162528378516	1.7486837471621484
5	41.0	1.0	1.9840365650382252	2.0	1.0	1.9840365650382252	1.9840365650382252	1.9840365650382252	1.0	1.0159634349617748
6	49.0	1.0	2.0	2.0	1.0	1.6787953743323004	2.0	2.0	1.0	1.3212046256676996
7	30.0	2.0	1.0	2.0	2.0	2.0	2.0	1.0	2.0	2.0
8	52.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
9	42.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	31.0	1.0	nan	1.0	2.0	2.0	2.0	2.0	2.0	2.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	50.0	1.0	1.0	2.0	1.0	1.9802867024699249	1.0197132975300751	2.0	2.0	1.0
2	54.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	1.0	1.4222208175933146
3	38.0	1.0	1.0	2.0	1.0	1.0	1.0034720026080026	2.0	1.0034720026080026	1.9965279973919974
4	59.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	1.9084887664587042
5	37.0	1.0	2.0	2.0	1.0454226275591819	2.0	2.0	2.0	2.0	2.0
6	61.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
7	33.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
8	39.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	2.0
9	25.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
10	38.0	1.0	2.0	1.0	1.0	1.0	1.0	2.0	2.0	2.0

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	47.0	1.0	1.0	2.0	1.0	1.0	1.2700531605029268	2.0	2.0	1.2700531605029268
2	42.0	1.0	1.930220704329241	2.0	1.0	1.930220704329241	1.930220704329241	1.930220704329241	1.0	1.069779295670759
3	52.0	1.0	1.0	2.0	1.0	1.0	1.5418377784160833	2.0	2.0	1.5418377784160833
4	45.0	1.0	1.3204448775255408	2.0	1.0	1.3204448775255408	2.0	2.0	1.6795551224744592	1.0
5	45.0	1.0	1.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0
6	45.0	1.0	1.6620812061576653	2.0	1.0	2.0	1.6620812061576653	2.0	2.0	1.0
7	30.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
8	45.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
9	36.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	33.0	1.0	1.292840245315713	2.0	1.0	1.707159754684287	1.707159754684287	2.0	1.707159754684287	1.292840245315713
2	57.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0
3	54.0	1.0	2.0	2.0	1.0	1.7364646649353412	1.7364646649353412	1.2635353350646588	1.2635353350646588	1.2635353350646588
4	54.0	1.0	1.0	2.0	1.0	1.0	1.1281425832472314	1.1281425832472314	1.0	2.0
5	41.0	1.0	1.9249323561835276	2.0	1.0	2.0	1.9249323561835276	2.0	1.0750676438164724	1.0
6	42.0	1.0	1.9260488722254387	2.0	1.0	1.9260488722254387	1.9260488722254387	2.0	2.0	1.0
7	23.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	2.0	2.0
8	22.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0
9	51.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	1.0	1.0
10	40.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	2.0

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	42.0	1.0	1.0	2.0	1.0	1.0	1.6006895512852024	1.0	1.0	2.0
2	48.0	1.0	1.0301001495887525	2.0	1.0	1.0301001495887525	2.0	2.0	1.9698998504112475	1.0
3	43.0	1.0	1.6098036860429392	2.0	1.3901963139570608	1.3901963139570608	2.0	2.0	1.0	2.0
4	43.0	1.0	1.6657086827403165	2.0	1.0	1.6657086827403165	2.0	2.0	1.0	1.3342913172596835
5	50.0	1.0	1.0	2.0	1.0	1.0	1.7614567394951668	2.0	1.7614567394951668	1.2385432605048332
6	39.0	1.0	1.0657841554428554	1.0	1.0	1.0	2.0	2.0	1.0	2.0
7	38.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	1.0	2.0
8	51.0	1.0	1.0	2.0	1.0	2.0	1.0	2.0	2.0	1.0
9	30.0	1.0	2.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
10	34.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	47.0	1.0	1.0927642415811203	2.0	1.0	1.0	1.9072357584188797	2.0	1.0927642415811203	2.0
2	41.0	1.0	1.0	2.0	1.0	1.0466860917285392	1.0466860917285392	2.0	2.0	1.0
3	46.0	1.0	1.9832156313852907	2.0	1.0	1.0	2.0	2.0	1.0167843686147093	1.9832156313852907
4	45.0	1.0	1.0	2.0	1.0	1.0	1.4821117130881984	1.0	1.0	2.0
5	31.0	1.0	1.0124590943762926	2.0	1.0	1.0	1.0	2.0	2.0	1.0124590943762926
6	40.0	1.0	1.6525607594913525	2.0	1.0	1.0	1.0	2.0	2.0	1.6525607594913525
7	65.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	1.0	1.0
8	36.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
9	22.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0
10	66.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	39.0	1.0	1.3258972017169846	2.0	1.0	2.0	2.0	2.0	1.6741027982830154	1.0
2	46.0	1.0	1.5003993292901732	2.0	1.0	1.5003993292901732	2.0	1.0	1.0	1.4996006707098268
3	48.0	1.0	1.5169165931469046	2.0	1.0	1.5169165931469046	2.0	2.0	2.0	1.0
4	36.0	1.0	2.0	2.0	1.0	1.0	1.3991219567200437	2.0	1.0	2.0
5	53.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	1.5152827376898443
6	32.0	1.0	1.0	2.0	1.0	1.0	1.8876934491682729	2.0	2.0	1.8876934491682729
7	38.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	1.0	2.0
8	44.0	1.0	1.0	2.0	1.0	1.0	2.0	1.0	1.0	2.0
9	23.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
10	28.0	1.0	2.0	2.0	1.0	1.0	2.0	2.0	2.0	2.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	Age	Sex	Steroid	Antivirals	Fatigue	Malaise	Anorexia	Liver Big	Liver Firm	Spleen Palpable
1	57.0	1.0	1.2933682811069023	2.0	1.2933682811069023	1.2933682811069023	2.0	2.0	2.0	2.0
2	30.0	1.0	1.8239095567717052	2.0	1.0	1.1760904432282948	1.1760904432282948	2.0	1.1760904432282948	1.8239095567717052
3	52.0	1.0	1.0	2.0	1.0	1.0958922604848516	2.0	2.0	2.0	1.0
4	40.0	1.0	2.0	2.0	1.0	2.0	2.0	2.0	2.0	1.3997183015927361
5	46.0	1.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	1.0
6	57.0	1.0	1.3157325198646443	2.0	1.3157325198646443	1.3157325198646443	2.0	2.0	2.0	2.0
7	56.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	2.0
8	52.0	1.0	1.0	1.0	2.0	2.0	2.0	2.0	2.0	2.0
9	28.0	1.0	1.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0
10	49.0	1.0	1.0	1.0	1.0	1.0	1.0	2.0	1.0	2.0